TEXORA · LECTURE SHEET

Definition and Classification of Technical Textiles

Textiles chosen for what they do, not how they look: a definition of technical textiles, the line between them and conventional textiles, the standard twelve-area classification by end use, and — at principle level — why this sector keeps growing.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 1 · LEVEL 4 TERM 2

1. Definition

A widely used definition, from the *Textile Terms and Definitions* tradition, states: **technical textiles are textile materials and products manufactured primarily for their technical performance and functional properties rather than their aesthetic or decorative characteristics.** Every word of that definition matters:

- **Manufactured primarily for performance** — the product is specified by numbers first: breaking force, filtration efficiency, permeability, thermal resistance, chemical resistance. A geotextile is bought to a specification sheet, not to a shade card.
- **Materials *and* products** — the term covers fibres, yarns, fabrics and finished articles alike: a carbon fibre tow, a woven filter cloth and a finished airbag are all technical textiles.
- **Rather than aesthetics** — appearance is not irrelevant (a car interior technical textile must also look right), but it is never the primary reason for the specification.

The older term **industrial textiles** is narrower and is now avoided as the main label: much of the sector serves medicine, agriculture, sport and personal protection — not “industry” in the factory sense. “Technical textiles” is the accepted umbrella term; **performance textiles** and **functional textiles** are near-synonyms used for particular families.

2. Technical versus conventional textiles

ASPECT	CONVENTIONAL TEXTILES	TECHNICAL TEXTILES
Primary buying criterion	Appearance, handle, fashion, comfort	Measured functional performance against a specification

Typical specification	Colour, design, GSM range	Strength, permeability, pore size, thermal or chemical resistance, service life
Fibres used	Mostly cotton, polyester, viscose, wool	Those, plus high-performance fibres: aramid, glass, carbon, HMPE, PTFE, high-tenacity polyester
Structures	Woven, knitted apparel and household fabrics	Woven, knitted, braided, nonwoven, 3D and composite structures chosen by function
Testing emphasis	Colour fastness, shrinkage, appearance	Application-specific performance tests, often regulated (e.g. protective clothing standards)
Price logic	Cost per metre / per garment	Cost per unit of performance over service life

Table 1 — The contrast is one of emphasis, not of kind: the same mill may weave both, but the design brief is written differently.

> **A USEFUL TEST**
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> Ask of any product: *if it failed, what exactly would fail — the look, or the function?* A faded curtain has failed aesthetically; a burst filter bag, a torn seat belt or a geotextile that lets a road subgrade pump mud has failed functionally. Products whose failure is functional, and which are engineered against that failure, are technical textiles.

3. Classification by end use — the twelve areas

The standard classification used across the industry (popularised by the Techtextil exhibition and David Rigby Associates) groups technical textiles into **twelve application areas**, each with a short name ending in *-tech*:

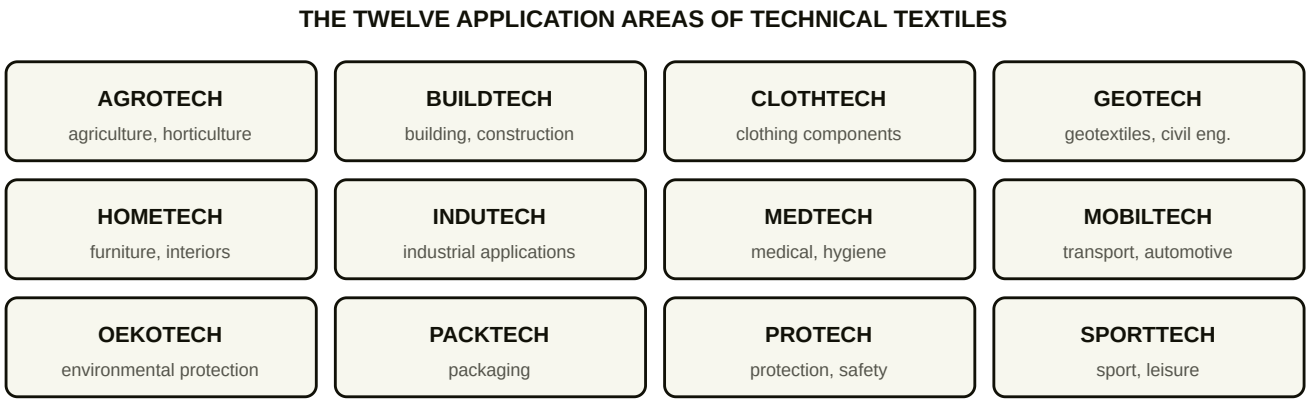


Figure 1 — The twelve-area (Techtextil/DRA) classification of technical textiles by application.

AREA	COVERS	TYPICAL PRODUCTS
Agrotech	Agriculture, horticulture, forestry, fishing	Shade nets, crop covers, mulch mats, fishing nets, ropes
Buildtech	Building and architectural construction	Membrane roofs, scaffolding nets, insulation, concrete reinforcement textiles
Clothtech	Functional components of clothing and footwear	Interlinings, sewing threads, linings, shoe components, zips and tapes
Geotech	Civil engineering and earthworks	Geotextiles and geogrids for separation, filtration, drainage, reinforcement
Hometech	Furniture, interior furnishing, floor coverings	Upholstery components, mattress ticking and springs covers, carpets, blinds
Indutech	Industrial products and processes	Filter fabrics, conveyor belts, abrasives backing, gaskets, polishing cloths
Medtech	Medical and hygiene products	Wound dressings, sutures, implants, surgical gowns, hygiene nonwovens
Mobiltech	Automobiles, rail, ships, aircraft	Seat belts, airbags, tyre cord, upholstery, insulation, composites
Oekotech	Environmental protection	Oil-absorbent booms, landfill liners support, erosion control, air filtration

Packtech	Packaging	Woven sacks, bulk bags (FIBC), nets, tea bags, protective sleeves
Protech	Personal and property protection	Flame-retardant and ballistic clothing, cut-resistant gloves, fall-arrest webbing
Sporttech	Sport and active leisure	Parachutes, sails, tents, sports surfaces, performance apparel components

Table 2 — The twelve areas with typical products. Topic 4 of this course examines several of them in detail.

Other classifications exist and are useful for other questions: by **structure** (woven, knitted, braided, nonwoven, composite), by **fibre type** (natural, regenerated, synthetic, high-performance inorganic), and by **function** (mechanical, fluid handling, thermal, electrical, protective). But the twelve-area end-use scheme is the one the market itself uses, so it is the primary scheme for this course.

4. Where technical textiles happen in the value chain

A technical textile may enter at any stage. **Fibre-level** products are sold as performance raw materials (high-tenacity yarn, tyre cord). **Fabric-level** products are engineered roll goods (filter cloth, coated fabric). **Component-level** products are parts inside another product (airbag module fabric, composite preform). **Finished articles** are complete goods (a tent, a surgical drape set). The further the product sits from the fibre, the more the specification is written by the end industry — automotive, medical, civil engineering — rather than by the textile mill, which is why technical textile businesses must understand their customers' industries, not only textiles.

5. Market overview and growth — the principles

Market reports differ in coverage and method, and this course does not rest on any particular report's numbers. What can be stated reliably, as structural principles, is *why* demand for technical textiles grows even when apparel demand is flat:

- **Infrastructure investment.** Roads, railways, dams and land reclamation consume geotextiles and buildtech products in proportion to construction activity; every kilometre of road on soft soil is a potential geotextile specification.
- **Vehicle production and lightweighting.** Each vehicle carries a substantial mass of textiles (seat belts, airbags, upholstery, insulation, tyre reinforcement), and replacing metal with textile-reinforced composites saves weight and fuel — a permanent engineering pressure, not a fashion cycle.

- **Healthcare growth and ageing populations.** Hygiene nonwovens, wound care and implantable textiles grow with healthcare spending and with the number of surgical procedures.
- **Regulation.** Safety law creates technical textile demand directly: protective clothing standards, fire regulations for public buildings and transport, filtration and emission rules. When a standard is tightened, a market is created or enlarged.
- **Substitution of heavier materials.** Textiles replace metal, wood and heavier plastics where flexibility, corrosion resistance, or weight matters — hoses, belts, ropes, composite panels.
- **Environmental pressure.** Filtration, erosion control and water treatment (Oekotech/Geotech) grow with environmental regulation and climate adaptation spending.

> **HOW TO READ ANY MARKET CLAIM**
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> When a source quotes a market size or growth rate, ask four questions: (1) Does it count fibres only, fabrics only, or finished products too? (2) Which countries does it cover? (3) Is growth quoted in volume (tonnes) or value (currency)? — the two move differently because technical products raise in value per kilogram. (4) What base year and forecast method? A claim that survives these four questions is usable; one that cannot answer them is advertising.

For a country like Bangladesh, whose textile industry is apparel-led, the principle-level opportunity is diversification: technical textiles offer product families whose demand follows infrastructure, health and industry rather than fashion seasons — but they demand specification selling, testing capability and certification, which is the real barrier to entry, not machinery alone.

Glossary

Technical textile — textile material or product made primarily for technical performance and functional properties rather than aesthetic characteristics. **High-performance fibre** — fibre engineered for exceptional strength, heat resistance, chemical resistance or modulus (aramid, carbon, glass, HMPE). **Specification** — the documented set of measurable properties a product must meet. **Geotextile** — permeable textile used in contact with soil in civil engineering. **Nonwoven** — fabric made directly from a web of fibres bonded mechanically, thermally or chemically, without yarn formation.

Key points

• Definition: performance and function first, aesthetics second — across fibres, fabrics and finished products.

• The twelve areas (Agrotech…Sporttech) classify by end use; the same technology can serve several areas.

• Technical textiles are specified numerically and often regulated; the buying industry writes the real specification.
• Growth is driven structurally by infrastructure, vehicles, healthcare, regulation, substitution and environmental needs.
• Judge market figures by coverage, geography, volume-vs-value and method — never quote a number you cannot interrogate.

Practice

1. Classify each product into its twelve-area category and justify in one line: (a) a woven polyester fabric used as a road-separation layer; (b) a needle-punched nonwoven used inside a car door as sound insulation; (c) an aramid fabric in a firefighter's jacket; (d) a polypropylene woven sack for fertiliser; (e) a knitted spacer fabric used as a wound dressing.
2. A buyer says: “This curtain fabric is a technical textile because we test its flame retardancy.” Assess the claim using the definition in Section 1.
3. Explain why two market reports can give very different sizes for “the technical textiles market” while both are honestly compiled.

Answer key

1. (a) **Geotech** — designed and sold for a civil-engineering function in soil. (b) **Mobiltech** — a transport application; the end use, not the nonwoven structure, decides. (c) **Protech** — personal protection against heat and flame. (d) **Packtech** — packaging. (e) **Medtech** — a medical product, even though the same spacer structure could be Hometech elsewhere.
2. Partly right: flame retardancy is a functional, tested property, so if the fabric is *specified and bought primarily* for certified fire performance (e.g. for a public building under fire regulation), it sits in technical textiles (Buildtech/Hometech boundary, decided by end use). If the retardancy is only a secondary quality of a decorative curtain, the product remains conventional — the definition turns on the *primary* purpose.
3. They may count different things (fibres vs fabrics vs finished goods), cover different regions, measure volume instead of value, or include/exclude borderline families such as hygiene nonwovens. Honest compilers using different coverage will produce different totals; the figures are only comparable when coverage, geography and measure are matched.